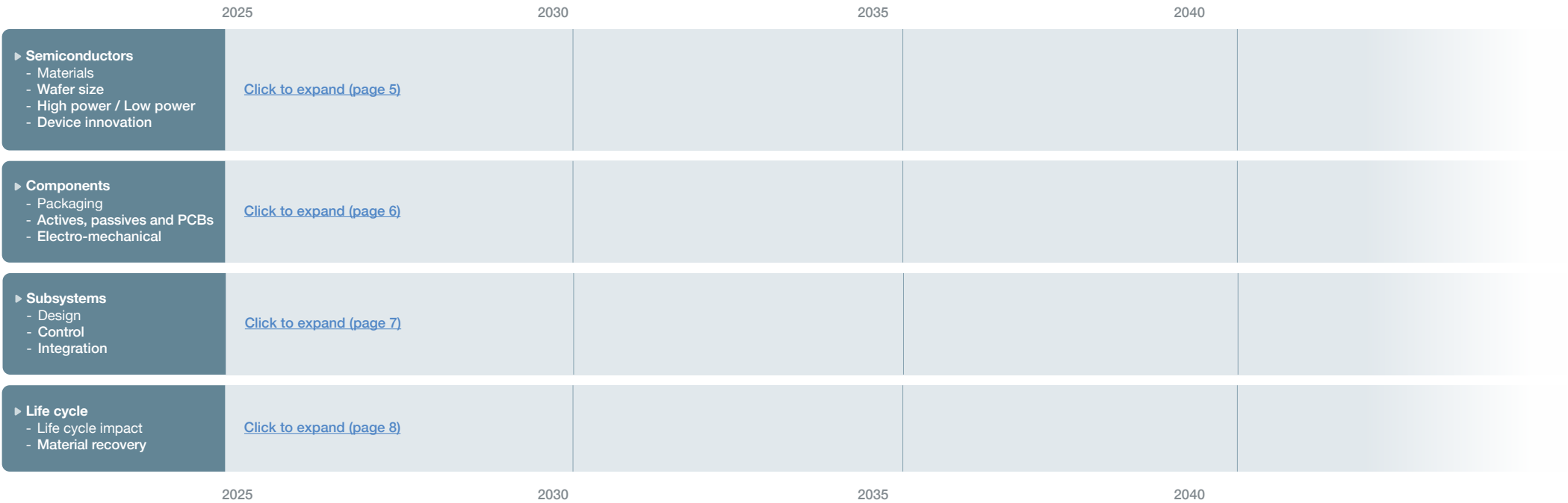


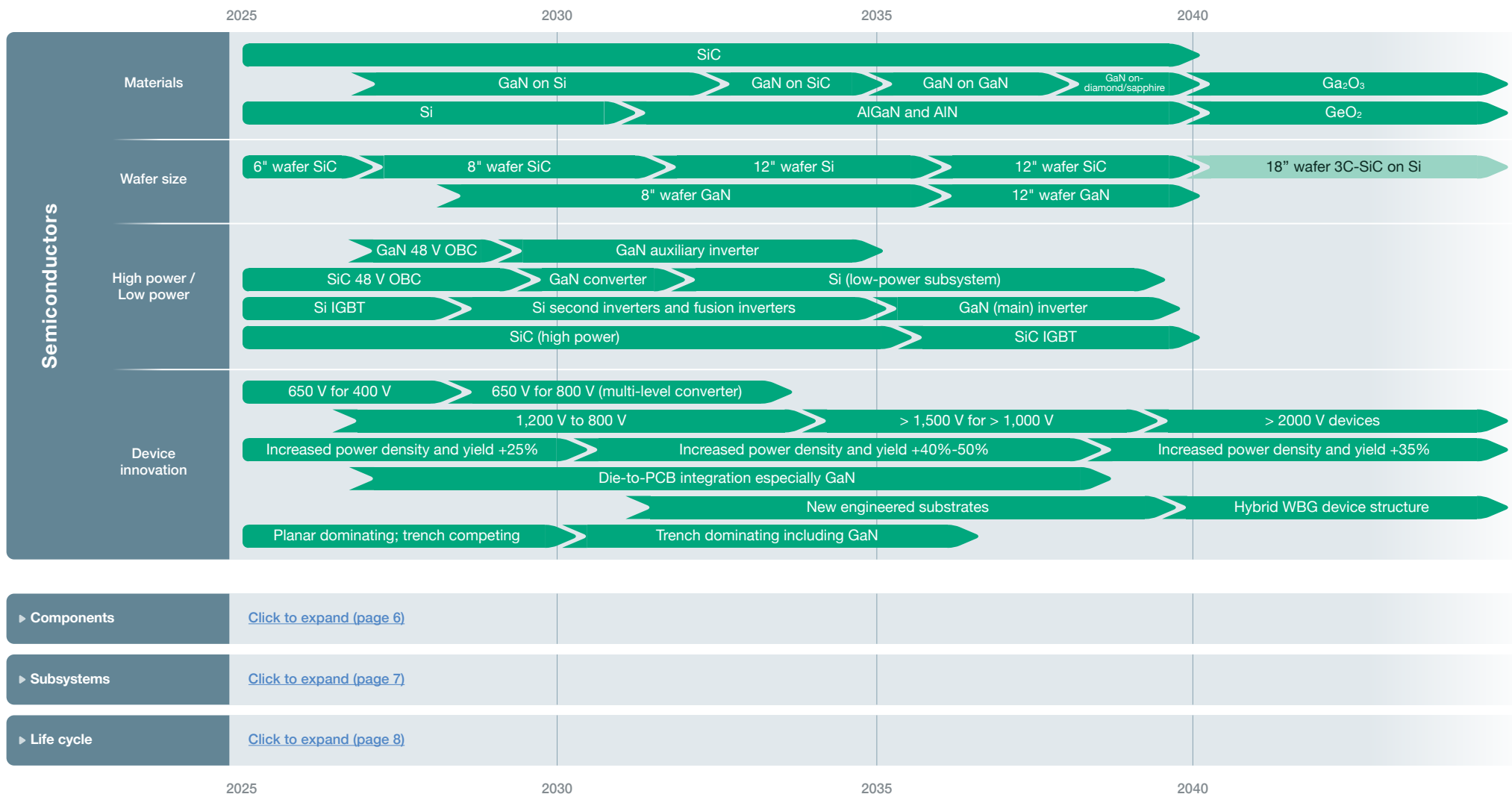


Technology is in a mass market application. Significant innovation is expected in this timeframe.

Transitions do not mean a phase-out from market but a change of R&D emphasis.

Fluid timings: these technologies have less consensus on when they will occur on the timeline, and may be implemented earlier or later than they appear. They may be adopted in niche vehicle applications.

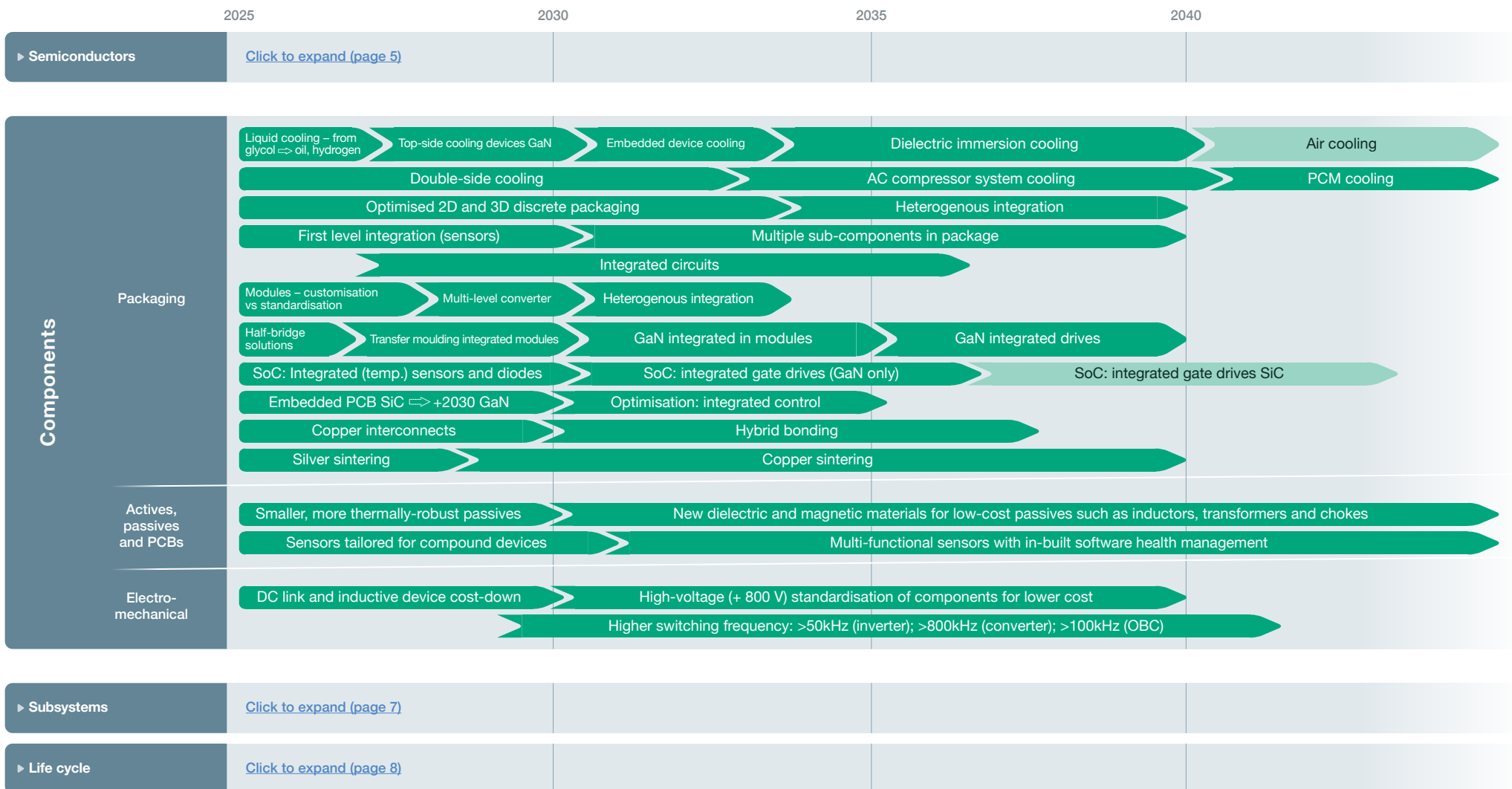
This roadmap represents a snapshot-in-time view of the global automotive industry propulsion technology forecast for mass market adoption. Specific application-tailored technologies will vary from region to region.



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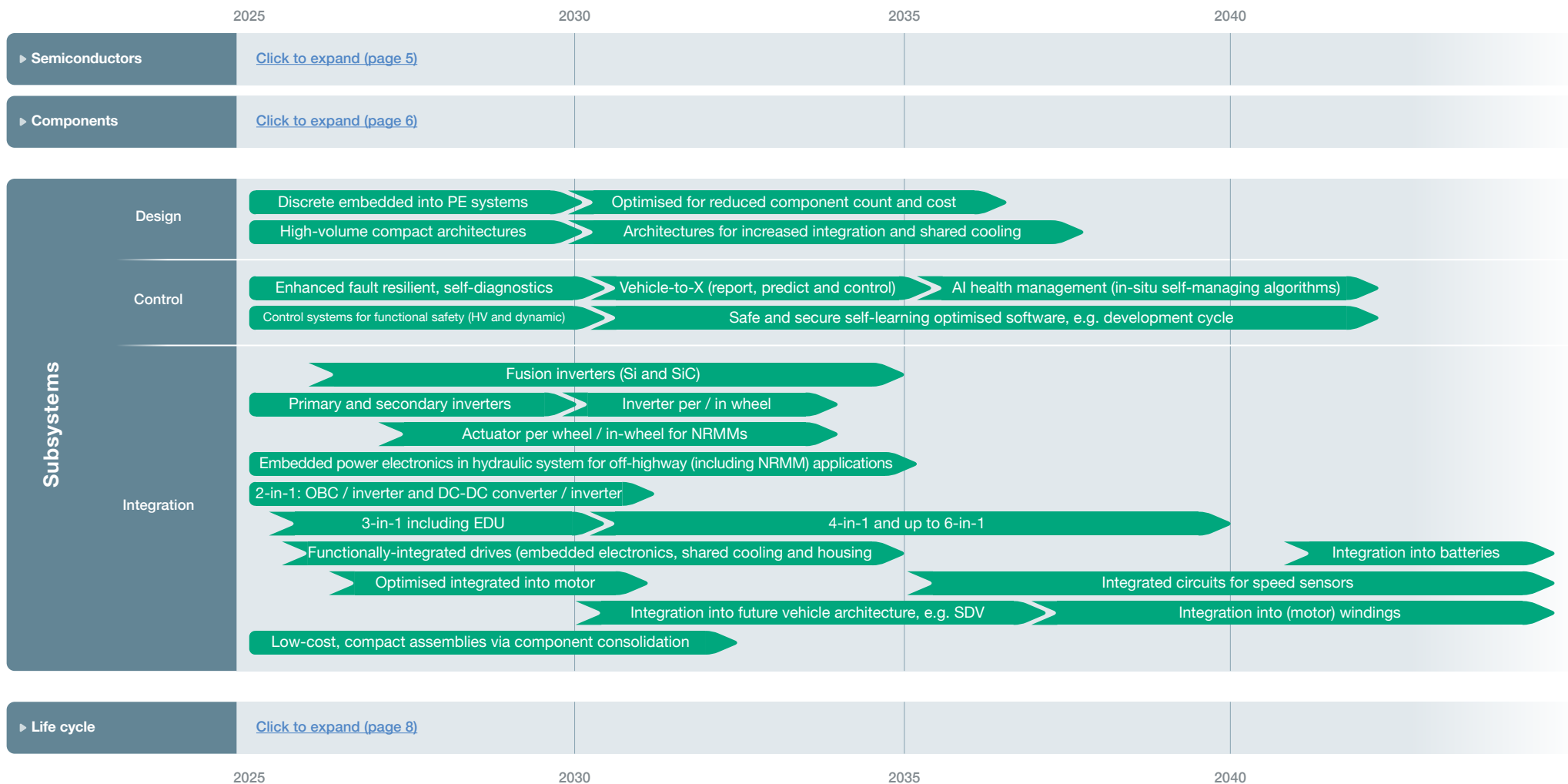
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Technology indicators for INVERTERS for light-duty vehicles

Technology indicators that industry is likely to achieve in a mass-market competitive environment.

All the performance metrics are ambitious and dependant on technology development but relate to the same technology.

	Light-duty vehicles	Urban mobility	Mass volume	Luxury / SUV	Performance
2025	Peak power (kW)	<100	250	350	600*
	Continuous power (kW)	50	150	230	450-600
	Input voltage (max) (V)	400	400/800	800	800
	Output current (A rms)	300	450/550	850	1,000
	Coolant inlet temp. (°C)	65	65	65	75
	Production volume	>100k	>100k	>100k	<10k

*Single inverter, multi-level; multi-phased system

	Light-duty vehicles	Urban mobility	Mass volume	Luxury / SUV	Performance
2030	Peak power (kW)	<150	250	500*	800*
	Continuous power (kW)	75	150	350-375	650-800
	Input voltage (max) (V)	400	800	800	1,200
	Output current (A rms)	200-350	550	1,000	1,500
	Coolant inlet temp. (°C)	65	65	75	85
	Production volume	>100k	>100k	>100k	<10k

	Light-duty vehicles	Urban mobility	Mass volume	Luxury / SUV	Performance
2040	Peak power (kW)	150	250	1,400	1,500
	Continuous power (kW)	90	150	800	1.0MW – 1.5MW
	Input voltage (max) (V)	400	400/800	1,200	1,500
	Output current (A rms)	350	450/550	1,000	1,500
	Coolant inlet temp. (°C)	65	75	85	Air cooled
	Production volume	>100k	>100k	>100k	<10k

		2025	2030	2040
All LDVs	Volumetric density(kW/l)	60-80	120	200
	Gravimetric density (kW/kg)	60-80	120	200
	WLTP** average efficiency	98%	98.5%	99%

Continuous power = 30 minutes

*Single inverter, multi-level; multi-phased system

**Worldwide Harmonised Light-Vehicle Test Procedure

Technology indicators for DC-DC CONVERTERS for light-duty vehicles

Technology indicators that industry is likely to achieve in a mass-market competitive environment.

All the performance metrics are ambitious and dependant on technology development but relate to the same technology.

	Light-duty vehicles	Urban Utility	Mass volume	Luxury / SUV	Performance
2025	Peak power (kW)	<2	>3	3.5	4
	Continuous power (kW)	<2	>3	3.5	2.5
	Input / output voltage (nominal) (V)	400	400	800	800
	Output / input voltage (nominal) (V)	12	12/48	12/48	12/48
	Coolant inlet temp. (°C)	65	65	65	75
	Production volume	>100k	>100k	>100k	<10k

	Light-duty vehicles	Urban Utility	Mass volume	Luxury / SUV	Performance
2030	Peak power (kW)	2	>3	4	5.2
	Continuous power (kW)	2	>3	4	5.2
	Input / output voltage (nominal) (V)	400	800	800	1,200
	Output / input voltage (nominal) (V)	12	12/48	48	48
	Coolant inlet temp. (°C)	65	65	75	85
	Production volume	>100k	>100k	>100k	<10k

Continuous power = 30 minutes

	Light-duty vehicles	Urban Utility	Mass volume	Luxury / SUV	Performance
2040	Peak power (kW)	3	>3	4	6.5
	Continuous power (kW)	3	>3	4	6.5
	Input / output voltage (nominal) (V)	400	400/800	800	1,500
	Output / input voltage (nominal) (V)	12	12/48	48	85
	Coolant inlet temp. (°C)	65	75	85	air cooled
	Production volume	>100k	>100k	>100k	<10k

		2025	2030	2040
All LDVs	Volumetric density(kW/l)	3 – 4	3 – 5	3 – 6
	Gravimetric density (kW/kg)	2 – 2.75	2.5-4	2.5-4.5
	WLTP** average efficiency	98%	98.50%	>99%

Continuous power = 30 minutes

**Worldwide Harmonised Light-Vehicle Test Procedure



Technology indicators for ON-BOARD CHARGERS for light-duty vehicles

Technology indicators that industry is likely to achieve in a mass-market competitive environment.

All the performance metrics are ambitious and dependant on technology development but relate to the same technology.

	Light-duty vehicles	Urban Utility	Mass volume	Luxury / SUV	Performance
2025	Peak power (kW)	~3	~7	11	11
	Continuous power (kW)	~3	~7	11	11
	Output / input voltage (nominal) (V)	400	400/800	800	800
	Coolant inlet temp. (°C)	65	65	65	75
	Production volume	>100k	>100k	>100k	<10k

	Light-duty vehicles	Urban Utility	Mass volume	Luxury / SUV	Performance
2030	Peak power (kW)	~3	11	22	22
	Continuous power (kW)	~3	11	22	22
	Output / input voltage (nominal) (V)	400	800	800	1,200
	Coolant inlet temp. (°C)	65	65	75	85
	Production volume	>100k	>100k	>100k	<10k

	Light-duty vehicles	Urban Utility	Mass volume	Luxury / SUV	Performance
2040	Peak power (kW)	~3	22	22	22
	Continuous power (kW)	~3	22	22	22
	Output / input voltage (nominal) (V)	400	400/800	800	1,500
	Coolant inlet temp. (°C)	65	75	85	air cooled
	Production volume	>100k	>100k	>100k	<10k

		2025	2030	2040
All LDVs	Volumetric density(kW/l)	2.5 – 3 n/a*	>4 n/a*	5 n/a*
	Gravimetric density (kW/kg)	3 – 3.5 4.5*	4.5 12*	5.5 <20*
	WLTP** average efficiency	98%	98.50%	99%

*Applies to performance cars

** WLTP = Worldwide Harmonised Light-Vehicle Test Procedure

Technology indicators for INVERTERS for heavy-duty vehicles

Technology indicators that industry is likely to achieve in a mass-market competitive environment.

All the performance metrics are ambitious and dependant on technology development but relate to the same technology.

Heavy-duty vehicle		HDV 7.5 t - 26 t HDV 26 t – 44 t*	
2025	Peak power (kW)	280-300	400*
	Continuous power (kW)	100-180	200*
	System voltage (V)	800	800
	Charging voltage (V)	650	750
	Output current (A rms)	750	820
	Coolant inlet temp. (°C)	65	65

Heavy-duty vehicle		HDV 7.5 t - 26 t HDV 26 t – 44 t*	
2030	Peak power (kW)	300	400*
	Continuous power (kW)	180	250*
	System voltage (V)	800	800
	Charging voltage (V)	750	1,200
	Output current (A rms)	750	820
	Coolant inlet temp. (°C)	65	65

Heavy-duty vehicle		HDV 7.5 t - 26 t HDV 26 t – 44 t*	
2040	Peak power (kW)	400	500*
	Continuous power (kW)	250	300*
	System voltage (V)	1,000	1,000
	Charging voltage (V)	1,200	1,800
	Output current (A rms)	820	1,000
	Coolant inlet temp. (°C)	75	75

*This vehicle typically has two motors. The output is per motor.

		2025	2030	2040
Heavy-duty 7.5 t - 26 t	Volumetric density(kW/l)	18.3	20.7	36.7
	Gravimetric density (kW/kg)	15	18.2	22
	WLTP** average efficiency	>97%	>98%	>98%

		2025	2030	2040
Heavy-duty 26 t - 44 t	Volumetric density(kW/l)	36.7	36.7	40.0
	Gravimetric density (kW/kg)	22.0	22.0	30.0
	WLTP** average efficiency	>97%	>98%	>98%

**Worldwide Harmonised Light-Vehicle Test Procedure